

Avni Singh

Applied AI Engineer

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SUMMARY

Applied AI Engineer with 1+ year of industry experience building AI-powered software and production-grade backend systems — LLM applications, RAG pipelines, AI agents, scalable REST APIs and full-stack systems. Strong in Python, FastAPI and distributed systems.

EXPERIENCE

AI/ML Intern — Bharat Forge, Pune

Jul 2025 – Jul 2026

- **Fleet Management System** — Backend REST APIs and graph-based task scheduling over MongoDB, MQTT and Express, with React dashboards for live fleet telemetry.
- **Warehouse Management System** — QR-based task creation across React, Node, Express and PostgreSQL, integrated with adjacent internal systems.
- Robotics foundations (ROS 2), VLA research, modular architecture and multi-agent path finding (MAPF) for multi-robot coordination.

Research Intern — DRDO, Mussoorie

Dec 2024 – Jan 2025

- Comparative study of AI/ML-enabled air defence systems and their operational trade-offs.

PROJECTS

- **Interview Coach AI (Jul 2026 – Ongoing)** — Multi-agent mock interview system with interviewer, evaluator and router agents in a stateful loop; RAG over ChromaDB, MCP web search, guardrails, LLM-as-judge evals. LangGraph, LangChain, FastAPI, Python.
- **MultiModal Vision-Language-Action Framework (Aug – Nov 2025)** — Explored VLA models combining visual perception, language understanding and action generation. PyTorch, HuggingFace Transformers, Computer Vision.
- **DeepSea Vision (Mar – May 2025)** — SRGAN-based underwater image enhancement with adversarial and perceptual learning, served via FastAPI.
- **PlotPoint (Aug – Oct 2024)** — AI pipeline for automated movie script and scene generation. GPT-2, LoRA, RAG, FAISS, Stable Diffusion, Gemini-Pro.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Applied AI: LLMs, RAG, LangChain, LangGraph, AI Agents, Multi-Agent Systems, MCP, LLM Evals, Guardrails, Multimodal AI, Computer Vision, Generative AI, Prompt Engineering

Deep Learning: PyTorch, Transformers, Hugging Face, OpenCV, YOLO, GANs, Diffusion Models

Backend & AI Infra: FastAPI, REST APIs, Model Inference, FAISS, ChromaDB, Vector Databases

Databases & Tools: PostgreSQL, MongoDB, Git, Docker, Linux

Computer Science: DSA, Operating Systems, Computer Networks, Distributed Systems, Concurrency

Cloud: AWS

EDUCATION & RECOGNITION

B.Tech, Artificial Intelligence & Machine Learning — Symbiosis Institute of Technology, Aug 2022 – May 2026 · CGPA 7.8

- 1 of 7 winners from 7,800+ applicants — Kalyani Group & WeCP Data, AI & Robotics Hackathon 2025.
- Publication: "LLM-based Abstractive Legal Document Summarization using Transformer Architectures", ICIRD 2024, Shinawatra University.

- Certifications: ChatGPT for Educators, Researchers & Professionals (World Technocon); GitHub Copilot Global Bootcamp (Microsoft Learn Student Ambassadors); What Is Generative AI? (LinkedIn); Introduction to Generative AI (Simplilearn).